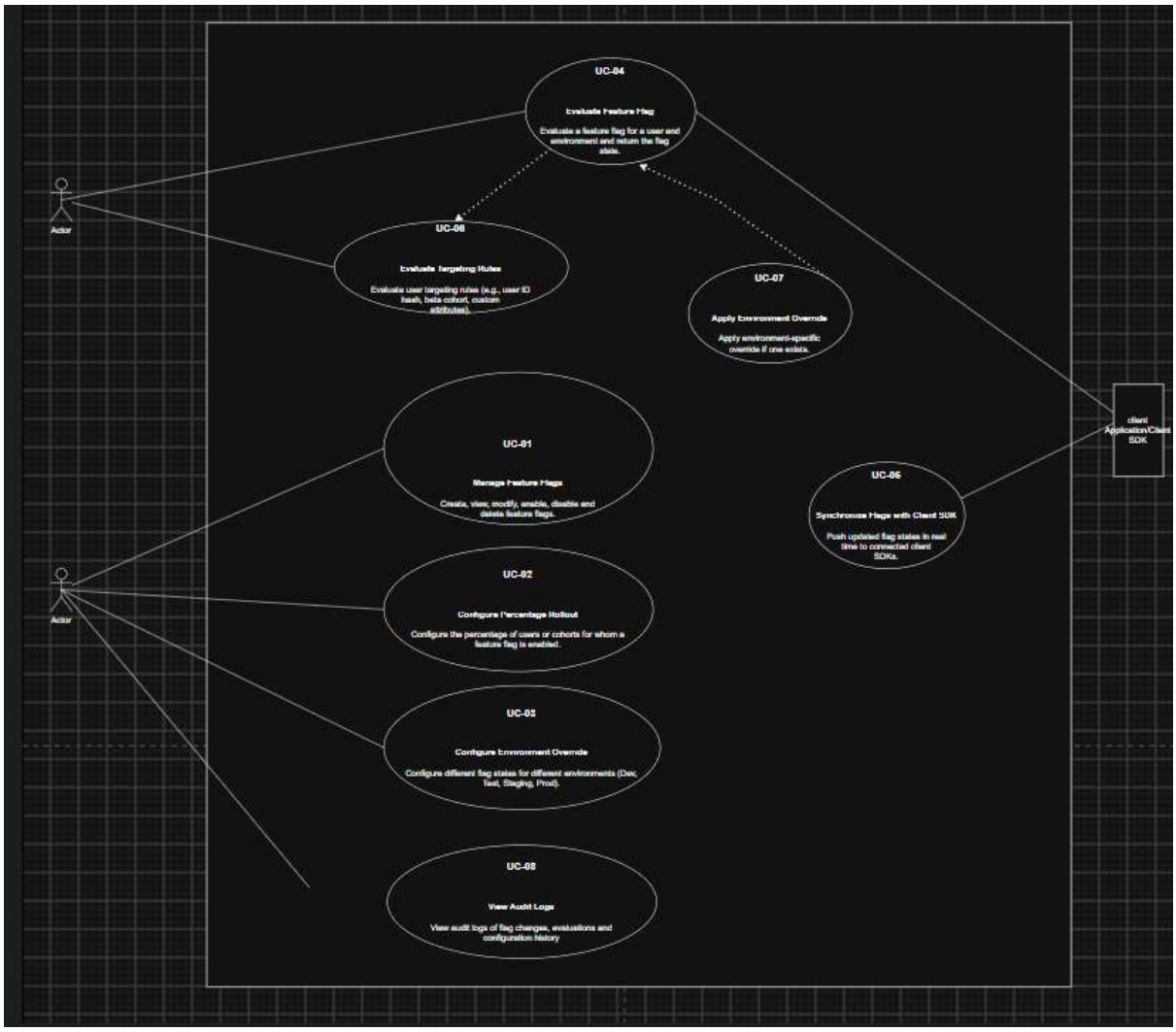
Lab 1 – Requirements Engineering

Problem Statement #43: Feature Flag & Dynamic Config Manager

Req ID	Type	Description	Priority	**Acceptance Criteria**	Rationale
FR-001	Functional	The system shall evaluate user targeting rules and return accurate boolean feature flag states.	High	**Pass:** A user in a configured 10% rollout receives the enabled flag. **Fail:** A user outside the cohort receives the enabled flag.	Ensures correct feature targeting.
FR-002	Functional	The system shall allow a Release Manager to configure the percentage of users for a feature rollout.	High	**Pass:** A configured 10% rollout assigns users according to the 10% cohort rule.	Supports gradual feature releases.
FR-003	Functional	The system shall allow a Release Manager to configure feature flag states for different environments.	High	**Pass:** A feature enabled in development and disabled in production returns the corresponding state for each environment.	Supports environment-specific configuration.

FR-004	Functional	The system shall allow a Release Manager to create, modify, and disable feature flags.	High	**Pass:** After changing a flag from disabled to enabled, subsequent evaluations return the updated state.	Provides centralized flag management.
FR-005	Functional	The system shall synchronize updated feature flag states with connected client SDKs.	High	**Pass:** A connected SDK receives an updated flag after the configuration is changed without requiring an application restart	Keeps client applications up to date.
NFR-001	NonFunctional	The feature flag evaluation API shall execute in under 15 ms.	High	**Pass:** Benchmarking confirms evaluation completes in less than 15 ms under the specified test load.	Minimizes performance overhead.
NFR-002	NonFunctional	The system shall synchronize updated feature configurations with connected client SDKs within 2 seconds.	High	**Pass:** At least 95 out of 100 configuration changes are delivered to connected SDKs within 2 seconds.	Ensures timely configuration updates.

UML Use-Case Diagram



Use-Case Flow Document

USE-CASE FLOW SPECIFICATION

Problem Statement #43

Feature Flag & Dynamic Config Manager

Use Case: UC-04 – Evaluate Feature Flag

Actors:

• Client Application / SDK

• Software Engineer

• Release Manager

Preconditions:

1. Feature flag is configured.

2. Targeting rules are available.

3. Client SDK is connected to the system.

Postconditions:

1. Feature flag state is determined.

2. Evaluated state is returned to the client.

Main Success Scenario:

1. Client SDK sends a feature flag evaluation request.

2. System identifies the requested feature flag.

3. System retrieves the flag configuration.

4. System evaluates the user's targeting rules.

5. System determines the user's rollout cohort.

6. System checks the applicable environment configuration.

7. System determines the final flag state.

8. System returns the flag state to the client SDK.

9. Client SDK enables or disables the feature accordingly.

10. Use case ends successfully.

Alternate Flow – Environment Override:

6a. System detects an environment-specific override.

6b. System applies the environment override.

6c. System returns the overridden flag state to the client SDK.

6d. Use case ends successfully.